

THE DEGREE-DIFFERENCE PRINCIPLE AND AFFINE SLICES OF BINARY-FORM FACTORISATION SPACES

OPENAI CODEX / ANTHROPIC MODELS

ABSTRACT. Let W be a two-dimensional complex vector space and write $V_d = \text{Sym}^d(W^\vee)$. For positive integers r, s , consider the multiplication–resultant map

$$\Phi_{r,s} : V_r \times V_s \longrightarrow V_{r+s} \times \mathbb{A}^1, \quad (A, B) \longmapsto (AB, \text{Res}(A, B)).$$

We prove the determinant identity

$$\det D\Phi_{r,s} = (-1)^{s(r+1)}(r-s) \text{Res}(A, B)^2$$

and derive a degree-difference principle: when $r \neq s$, the resultant recovers the unique infinitesimal scaling direction forgotten by multiplication; the normalized factorisation space is a finite étale $\mu_{|s-r|}$ -torsor over the complement of the resultant and hyperplane divisors in $\mathbb{P}(V_r) \times \mathbb{P}(V_s)$; its generic degree is $|s-r|\binom{r+s}{r}$; and, when the two boundary divisors are prime, the complement has divisor class group $\mathbb{Z}/|s-r|\mathbb{Z}$.

We then classify all normalized linear–quadratic slices. For $0 \neq \ell \in V_3^\vee$, set

$$X_\ell = \{(L, Q) \in V_1 \times V_2 : \text{Res}(L, Q) = 1, \ell(LQ) = 1\}.$$

Let $\Gamma \subset \mathbb{P}(V_3)$ be the twisted cubic and $\mathcal{H}_\ell = \mathbb{P}(\ker \ell)$. We prove

$$X_\ell \simeq \mathbb{A}^3 \iff \mathcal{H}_\ell \text{ is tangent but nonosculating to } \Gamma.$$

More precisely, the three contact types $(1, 1, 1)$, $(2, 1)$ and (3) give Grothendieck classes $\mathbb{L}^3 - \mathbb{L}$, $\mathbb{L}^3$ and $\mathbb{L}^3 - \mathbb{L}^2$, respectively; in the osculating case $X_\ell \simeq \mathbb{G}_m \times \mathbb{A}^2$. The successful tangent slice is an iterated affine-line bundle and yields the marked-simple-root model of the three-dimensional Keller counterexample. Finally, the torsor argument shows that no untrimmed normalized linear–times–degree– $(d-1)$ slice can be $\mathbb{A}^d$ for $d \geq 4$.

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1. INTRODUCTION

The recently announced three-dimensional counterexample to the Jacobian conjecture admits a geometric reformulation in terms of multiplication of a binary linear form and a binary quadratic form. The construction separates three issues that look accidental in expanded

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coordinates: multiplication forgets a relative scaling; the resultant restores that missing local coordinate; and one particular affine hyperplane slice of the product space is itself affine three-space.

The public record contains several distinct contributions. Alpöge's announcement credited Fable for the explicit counterexample [1]. In the discussion hosted by Speyer, Will Sawin posted a ChatGPT-produced symmetric-product formulation prompted by Andy Jiang; Tao's account derived the explicit polynomial coordinates [5, 6]. Sawin later developed an iterated $\mathbb{A}^1$ -bundle explanation through further ChatGPT discussion, and Daniel Litt supplied a related bundle argument. Levinson gave a projective-geometric conic account of the tangent construction [5, 3]. Speyer also identified the vector bundle sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^1}^{\oplus 3} \longrightarrow \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2} \longrightarrow 0,$$

which explains the trivial projective-line bundle hidden in the construction [5]. In the same discussion, a commenter using the name Skooi recorded the $d - 2$ class-group obstruction and a marked-simple-root interpretation, while Shubhodip Mondal noted the broader $|r - s|$ pattern for two binary factors. Tao's discussion isolated three PGL_2 -orbits of hyperplanes and observed that the affine phenomenon occurs in the double-root orbit; a subsequent comment posed the converse classification explicitly [5, 6].

This paper gives a self-contained answer. Its first result is broader than the cubic problem.

Theorem 1.1 (Degree-difference principle). *Let $r, s \geq 1$ and choose the standard monomial coefficient orders on V_r, V_s, V_{r+s} . Then*

$$\det D\Phi_{r,s} = (-1)^{s(r+1)}(r-s) \mathrm{Res}(A, B)^2.$$

In particular, if $r \neq s$, the map $\Phi_{r,s}$ is étale on the coprime locus $\{\mathrm{Res} \neq 0\}$.

For $0 \neq \ell \in V_{r+s}^\vee$, put

$$\tilde{X}_{r,s,\ell} = \{(A, B) : \mathrm{Res}(A, B) = 1, \ell(AB) = 1\},$$

$$U_{r,s,\ell} = (\mathbb{P}(V_r) \times \mathbb{P}(V_s)) \setminus (\mathcal{R}_{r,s} \cup \mathcal{S}_{r,s,\ell}),$$

where $\mathcal{R}_{r,s}$ is the resultant divisor and $\mathcal{S}_{r,s,\ell} = \{\ell(AB) = 0\}$. If $r \neq s$, projectivisation is a finite étale $\mu_{|s-r|}$ -torsor

$$\tilde{X}_{r,s,\ell} \longrightarrow U_{r,s,\ell}.$$

The induced multiplication map has generic degree

$$|s - r| \binom{r + s}{r}.$$

If $\mathcal{R}_{r,s}$ and $\mathcal{S}_{r,s,\ell}$ are prime, then

$$\mathrm{Cl}(U_{r,s,\ell}) \simeq \mathbb{Z}/|s - r|\mathbb{Z}.$$

The same integer $s - r$ therefore controls the Jacobian determinant, the normalisation ambiguity, the generic covering degree and the divisor class group. The cubic construction is the first adjacent-degree case: $(r, s) = (1, 2)$, so the degree difference equals one and all four obstructions disappear simultaneously.

Our central classification is the following.

Theorem 1.2 (Complete cubic classification). *Let $0 \neq \ell \in V_3^\vee$, let*

$$\Gamma = \{[M^3] : [M] \in \mathbb{P}(V_1)\} \subset \mathbb{P}(V_3)$$

be the twisted cubic, and put $\mathcal{H}_\ell = \mathbb{P}(\ker \ell)$. Over $\mathbb{C}$,

$$\boxed{X_\ell \simeq \mathbb{A}^3 \iff \mathcal{H}_\ell \text{ is tangent but not osculating to } \Gamma.}$$

More precisely,

$\mathcal{H}_\ell \cap \Gamma$	geometry of X_ℓ	$[X_\ell] \in K_0(\text{Var}_{\mathbb{C}})$
three distinct points (1, 1, 1)	$X_\ell \not\simeq \mathbb{A}^3$	$\mathbb{L}^3 - \mathbb{L}$
tangent, nonosculating (2, 1)	$X_\ell \simeq \mathbb{A}^3$	$\mathbb{L}^3$
osculating (3)	$X_\ell \simeq \mathbb{G}_m \times \mathbb{A}^2$	$\mathbb{L}^3 - \mathbb{L}^2$

Equivalently, in the dual projective space, the successful functionals form the discriminant surface of binary cubics with the triple-root curve removed.

The tangent case gives a polynomial étale map $\mathbb{A}^3 \rightarrow \mathbb{A}^3$ whose fibre over a cubic is the set of its simple roots. Its generic degree is three, so it is not injective. In suitable coordinates it is exactly the announced Keller map; an explicit conjugacy and an exact verification are recorded in [Appendices A](#) and [B](#).

The degree-difference theorem also closes the most immediate higher-degree extension.

Corollary 1.3 (Uniqueness of the untrimmed linear-factor slice). *For $d \geq 4$ and every nonzero $\ell \in V_d^\vee$,*

$$\{(L, B) \in V_1 \times V_{d-1} : \text{Res}(L, B) = 1, \ell(LB) = 1\} \not\simeq \mathbb{A}^d.$$

Thus $d = 3$ is the unique degree at least three in which the untrimmed linear-times-degree- $(d-1)$ normalisation can be affine space.

The proof uses compactly supported Euler characteristic rather than only the class group. The normalized slice is a finite étale cover of degree $d-2$ of the boundary complement $U_{1,d-1,\ell}$; if it were $\mathbb{A}^d$, then $1 = (d-2)\chi_c(U)$, impossible for $d \geq 4$.

2. THE MULTIPLICATION-RESULTANT MAP

Let W be a two-dimensional complex vector space and $V_d = \text{Sym}^d(W^\vee)$. Fix coordinates X, Y on W and order the monomial basis of V_d as

$$X^d, X^{d-1}Y, \dots, Y^d.$$

For $A \in V_r$ and $B \in V_s$, write $R(A, B) = \text{Res}(A, B)$. We use the Sylvester convention in which the matrix has the s shifted coefficient rows of A followed by the r shifted coefficient rows of B . Thus $\text{Res}(X^r, Y^s) = 1$.

2.1. The lost tangent direction.

Lemma 2.1. *If $R(A, B) \neq 0$, then the kernel of the differential of multiplication $m(A, B) = AB$ is*

$$\ker dm_{(A,B)} = \mathbb{C}(A, -B).$$

Proof. Suppose $\dot{A}B + A\dot{B} = 0$. Since A and B are coprime, A divides $\dot{A}B$ and hence divides $\dot{A}$. Both are degree- r binary forms, so $\dot{A} = tA$ for some $t \in \mathbb{C}$. Substitution gives $\dot{B} = -tB$. The converse is immediate. $\square$

The kernel is the infinitesimal relative-scaling direction $(A, B) \mapsto (\lambda A, \lambda^{-1}B)$. The resultant has bidegree (s, r) :

$$R(\lambda A, \mu B) = \lambda^s \mu^r R(A, B).$$

Consequently

$$dR_{(A,B)}(A, -B) = (s - r)R(A, B). \tag{1}$$

This already proves étaleness on the coprime locus when $r \neq s$.

2.2. The determinant identity.

Theorem 2.2. *With the coefficient orders fixed above and the target ordered as the coefficients of AB followed by the resultant,*

$$\det D\Phi_{r,s} = (-1)^{s(r+1)}(r-s)R(A,B)^2.$$

For $r = s$, both sides vanish identically.

Proof. Assume first that $r \neq s$. By Lemma 2.1 and (1), the determinant is nonzero whenever $R(A,B) \neq 0$. The resultant hypersurface is irreducible: it is the image of the irreducible incidence variety of pairs of forms with a marked common root. Hence every irreducible factor of $\det D\Phi_{r,s}$ is a scalar multiple of R .

Under $(A,B) \mapsto (\lambda A, \mu B)$, the Jacobian determinant has bidegree $(2s, 2r)$. Indeed, the $r+s+1$ coefficients of AB each have weight $\lambda\mu$, the resultant has weight $\lambda^s\mu^r$, and the source volume form has weight $\lambda^{r+1}\mu^{s+1}$. Since R has bidegree (s, r) , there is a constant $c_{r,s}$ such that

$$\det D\Phi_{r,s} = c_{r,s}R^2.$$

Evaluate at $A = X^r$ and $B = Y^s$, where $R(A,B) = 1$. Write tangent coefficients as

$$\dot{A} = \sum_{i=0}^r a_i X^{r-i} Y^i, \quad \dot{B} = \sum_{j=0}^s b_j X^{s-j} Y^j.$$

The product differential sends b_j to the coefficient of $X^{r+s-j}Y^j$ and a_i to the coefficient of $X^{r-i}Y^{s+i}$. The only overlap is the middle coefficient, which is $a_0 + b_s$. The resultant differential is $sa_0 + rb_s$. Delete the two overlapping columns a_0, b_s and the corresponding product row. The remaining product rows form a permutation matrix. Moving the columns into block order has $s(r+1) + r$ inversions, while moving the resultant row next to the overlap row contributes r further inversions. Its total sign is therefore $(-1)^{s(r+1)}$. On the deleted coordinates and the product/resultant rows, the remaining block is

$$\begin{pmatrix} 1 & 1 \\ s & r \end{pmatrix},$$

whose determinant is $r - s$. Their product gives the displayed sign.

If $r = s$, (1) shows that the relative-scaling vector remains in the kernel of $D\Phi_{r,r}$ wherever $R \neq 0$, so the determinant vanishes identically. The formula then remains valid. $\square$

Remark 2.3. The determinant identity is integral. Over a field of characteristic p , the same proof shows that $\Phi_{r,s}$ is étale on the coprime locus exactly when $p \nmid (s - r)$. We work over $\mathbb{C}$ below because the orbit and Hodge-theoretic arguments are cleanest there.

2.3. Projective normalisation and the cyclic torsor.

Let

$$Y_{r,s} = \mathbb{P}(V_r) \times \mathbb{P}(V_s).$$

The resultant defines a prime divisor $\mathcal{R}_{r,s}$ of class (s, r) , and $\ell(AB)$ defines a divisor $\mathcal{S}_{r,s,\ell}$ of class $(1, 1)$.

Proposition 2.4. *Let $n = |s - r| > 0$. Projectivisation induces a finite étale μ_n -torsor*

$$q : \tilde{X}_{r,s,\ell} \longrightarrow U_{r,s,\ell}.$$

The action is

$$\zeta \cdot (A, B) = (\zeta A, \zeta^{-1} B), \quad \zeta^n = 1.$$

Proof. Take $([A], [B]) \in U_{r,s,\ell}$ and representatives with

$$R_0 = R(A, B) \neq 0, \quad h_0 = \ell(AB) \neq 0.$$

We seek scalars λ, μ for which

$$\lambda^s \mu^r R_0 = 1, \quad \lambda \mu h_0 = 1.$$

The second equation gives $\mu = (\lambda h_0)^{-1}$. If $s > r$, the first becomes

$$\lambda^{s-r} = \frac{h_0^r}{R_0}. \quad (2)$$

If $r > s$, the equivalent equation is $\lambda^{r-s} = R_0/h_0^r$. Thus every projective point has exactly n normalized lifts, permuted simply transitively by μ_n .

On a product of standard affine charts of $Y_{r,s}$, choose regular representatives of the two tautological lines. The corresponding equation is locally a Kummer cover $t^n = u$ with u invertible. Over $\mathbb{C}$ this is finite étale. The local equations arise from the global projectivisation map and therefore agree on overlaps; étaleness is local on the source. The global μ_n -action is simply transitive on geometric fibres, so these local Kummer descriptions give the stated torsor. $\square$

Proposition 2.5 (Generic degree). *Assume $r \neq s$. The multiplication map*

$$\tilde{X}_{r,s,\ell} \longrightarrow \{C \in V_{r+s} : \ell(C) = 1\}$$

has generic degree

$$\boxed{|s - r| \binom{r+s}{r}}.$$

Proof. A general binary form C of degree $r+s$ has distinct roots. Choosing a degree- r factor projectively is equivalent to choosing r of those roots, giving $\binom{r+s}{r}$ projective factorizations. For each one, Proposition 2.4 gives $|s - r|$ normalized lifts. $\square$

2.4. Divisor classes and units.

Proposition 2.6. *Suppose $r \neq s$ and $\mathcal{S}_{r,s,\ell}$ is prime. Then*

$$\mathcal{O}(U_{r,s,\ell})^\times = \mathbb{C}^\times, \quad \text{Cl}(U_{r,s,\ell}) \simeq \mathbb{Z}/|s - r|\mathbb{Z}.$$

If $\mathcal{S}_{r,s,\ell}$ is reducible, then $\mathcal{O}(U_{r,s,\ell})^\times / \mathbb{C}^\times$ is nonzero.

Proof. For a smooth projective variety and the complement of prime divisors, the divisor localisation sequence gives

$$0 \longrightarrow \mathcal{O}(U)^\times / \mathbb{C}^\times \longrightarrow \bigoplus_i \mathbb{Z}[D_i] \longrightarrow \text{Pic}(Y_{r,s}) \longrightarrow \text{Cl}(U) \longrightarrow 0.$$

If the boundary has the two prime components $\mathcal{R}_{r,s}$ and $\mathcal{S}_{r,s,\ell}$, the class map is represented by

$$\begin{pmatrix} s & 1 \\ r & 1 \end{pmatrix}.$$

Its determinant is $s - r$, and its Smith normal form is $\text{diag}(1, |s - r|)$. The map is injective, so units are constant and the cokernel is $\mathbb{Z}/|s - r|\mathbb{Z}$.

If $\mathcal{S}_{r,s,\ell}$ is reducible, the boundary contains at least three prime divisors, while $\text{Pic}(Y_{r,s})$ has rank two. The class map has a nonzero kernel, which produces a nonconstant unit. $\square$

Corollary 2.7. *If $|s - r| > 1$, then*

$$\tilde{X}_{r,s,\ell} \not\cong \mathbb{A}^{r+s}.$$

Proof. After analytification, a finite étale morphism of degree n is an n -sheeted topological covering. Additivity over a finite constructible stratification by finite CW complexes therefore gives $\chi_c(X) = n\chi_c(Y)$. Applying this to [Proposition 2.4](#) yields

$$\chi_c(\tilde{X}_{r,s,\ell}) = |s - r| \chi_c(U_{r,s,\ell}).$$

If the source were $\mathbb{A}^{r+s}$, the left-hand side would equal one, which is not divisible by $|s - r| > 1$ in $\mathbb{Z}$. $\square$

Taking $(r, s) = (1, d - 1)$ proves [Corollary 1.3](#). Notice that this argument does not require either boundary divisor to be prime.

3. THE CUBIC COMPACTIFICATION AND MARKED ROOTS

We now specialize to $(r, s) = (1, 2)$. Write

$$X_\ell = \tilde{X}_{1,2,\ell}, \quad \overline{X} = \mathbb{P}(V_1) \times \mathbb{P}(V_2) \simeq \mathbb{P}^1 \times \mathbb{P}^2.$$

Since $s - r = 1$, [Proposition 2.4](#) has one sheet.

Proposition 3.1. *Projectivisation is an isomorphism*

$$X_\ell \xrightarrow{\sim} \overline{X} \setminus (\mathcal{R} \cup \mathcal{S}_\ell).$$

For representatives with $R_0 = \text{Res}(L, Q) \neq 0$ and $h_0 = \ell(LQ) \neq 0$, the inverse normalization is

$$L' = \frac{h_0}{R_0} L, \quad Q' = \frac{R_0}{h_0^2} Q.$$

Proof. The displayed scalings satisfy $\text{Res}(L', Q') = 1$ and $\ell(L'Q') = 1$, and they are unchanged if the original representatives are rescaled independently. They therefore define the inverse to projectivisation. $\square$

Identify $\mathbb{P}(V_d)$ with $\text{Sym}^d(\mathbb{P}^1)$. A point $[L] \in \mathbb{P}(V_1)$ is a root $p \in \mathbb{P}^1$, while $[Q] \in \mathbb{P}(V_2)$ is an unordered degree-two divisor D . The resultant divisor is the incidence locus

$$\mathcal{R} = \{(p, D) : p \in D\}.$$

The multiplication map

$$\pi : \mathbb{P}(V_1) \times \mathbb{P}(V_2) \longrightarrow \mathbb{P}(V_3), \quad ([L], [Q]) \longmapsto [LQ]$$

marks one root of a cubic divisor.

Proposition 3.2 (Marked-simple-root model). *There is a natural isomorphism*

$$X_\ell \simeq \{(C, p) : \ell(C) = 1, p \text{ is a simple root of } C\}.$$

Under this isomorphism, the map $f_\ell : X_\ell \rightarrow H_\ell = \{\ell(C) = 1\}$ forgets the marked root. Hence

$$\#f_\ell^{-1}(C) = \begin{cases} 3, & C \text{ has three distinct roots,} \\ 1, & C \text{ has one double and one simple root,} \\ 0, & C \text{ is a cube.} \end{cases}$$

Moreover, f_ℓ is étale.

Proof. The complement of $\mathcal{R}$ consists exactly of factorizations $C = LQ$ in which the root of L is not a root of Q , hence is a simple root of C . The condition $\ell(C) \neq 0$ removes $\mathcal{S}_\ell$, and [Proposition 3.1](#) imposes the unique affine normalisation. The fibre count follows immediately. Étaleness follows from [Theorem 2.2](#) by base change along $H_\ell \rightarrow V_3 \times \mathbb{A}^1$, $C \mapsto (C, 1)$. $\square$

4. HYPERPLANE CONTACT AND THE BOUNDARY DIVISORS

Let

$$\Gamma = \{[M^3] : [M] \in \mathbb{P}(V_1)\} \subset \mathbb{P}(V_3)$$

be the twisted cubic and $\mathcal{H}_\ell = \mathbb{P}(\ker \ell)$. The restriction of a hyperplane to $\Gamma \simeq \mathbb{P}^1$ has degree three. Over $\mathbb{C}$, the intersection divisor has one of the three types

$$(1, 1, 1), \quad (2, 1), \quad (3).$$

These are respectively the transverse, tangent nonosculating and osculating orbits.

For $p = [L] \in \mathbb{P}(V_1)$, denote the fibres of the two boundary divisors by

$$\mathcal{R}_p = \{[Q] : \text{Res}(L, Q) = 0\}, \quad (\mathcal{S}_\ell)_p = \{[Q] : \ell(LQ) = 0\}.$$

The first is always a line in $\mathbb{P}(V_2)$.

Lemma 4.1 (The missing bundle criterion). *Define*

$$\alpha_p : V_2 \longrightarrow \mathbb{C}, \quad Q \longmapsto \ell(LQ).$$

Then $\alpha_p = 0$ if and only if $\mathcal{H}_\ell$ is the osculating plane to Γ at $[L^3]$. Consequently, if $\mathcal{H}_\ell$ is not osculating, $\mathcal{S}_\ell \rightarrow \mathbb{P}^1$ is a $\mathbb{P}^1$ -bundle.

Proof. The functional α_p vanishes exactly when ℓ annihilates the three-dimensional subspace $LV_2 \subset V_3$, equivalently when

$$\mathcal{H}_\ell = \mathbb{P}(LV_2).$$

Choose a complementary linear form M . Expanding

$$(L + tM)^3 = L^3 + 3tL^2M + 3t^2LM^2 + t^3M^3$$

shows that the osculating plane at $[L^3]$ is

$$\mathbb{P}\text{Span}\{L^3, L^2M, LM^2\} = \mathbb{P}(LV_2).$$

If no α_p vanishes, their kernels form a rank-two subbundle of the trivial bundle $V_2 \otimes \mathcal{O}_{\mathbb{P}^1}$, whose projectivisation is $\mathcal{S}_\ell$. $\square$

Lemma 4.2. *One has $\mathcal{R}_p \subseteq (\mathcal{S}_\ell)_p$ if and only if $\mathcal{H}_\ell$ contains the tangent line to Γ at $[L^3]$. If $\alpha_p \neq 0$, so that both fibres are lines, this is equivalent to $\mathcal{R}_p = (\mathcal{S}_\ell)_p$.*

Proof. A quadratic form in $\mathcal{R}_p$ is $Q = LM$ for some $M \in V_1$. Its product with L is L^2M . As M varies, the projective line $\mathbb{P}(L^2V_1)$ is the tangent line to Γ at $[L^3]$. Thus $\mathcal{R}_p \subseteq (\mathcal{S}_\ell)_p$ exactly when that tangent line lies in $\mathcal{H}_\ell$. If $\alpha_p \neq 0$, both fibres are lines, and containment is equality. $\square$

Rescaling ℓ does not change the isomorphism type of X_ℓ : for $t \in \mathbb{C}^\times$, the map $(L, Q) \mapsto (tL, t^{-2}Q)$ identifies X_ℓ with $X_{t\ell}$. The SL_2 -action on binary forms acts equivariantly on the family. Hence it suffices to study one representative of each contact type.

5. MOTIVIC SEPARATION OF THE THREE ORBITS

Let $\mathbb{L} = [\mathbb{A}^1] \in K_0(\text{Var}_{\mathbb{C}})$. Since

$$[\overline{X}] = [\mathbb{P}^1][\mathbb{P}^2] = 1 + 2\mathbb{L} + 2\mathbb{L}^2 + \mathbb{L}^3,$$

and $\mathcal{R} \rightarrow \mathbb{P}^1$ is a $\mathbb{P}^1$ -bundle,

$$[\mathcal{R}] = (1 + \mathbb{L})^2 = 1 + 2\mathbb{L} + \mathbb{L}^2.$$

By Lemma 4.1, the same is true for $\mathcal{S}_\ell$ in the transverse and tangent nonosculating cases.

Proposition 5.1. *The three contact types have classes*

$$[X_\ell] = \begin{cases} \mathbb{L}^3 - \mathbb{L}, & (1, 1, 1), \\ \mathbb{L}^3, & (2, 1), \\ \mathbb{L}^3 - \mathbb{L}^2, & (3). \end{cases}$$

Proof. In the transverse case, Lemma 4.2 shows that the two fibre lines are distinct for every p . Their intersection is therefore a section of $\overline{X} \rightarrow \mathbb{P}^1$, so $[\mathcal{R} \cap \mathcal{S}_\ell] = 1 + \mathbb{L}$. Inclusion–exclusion gives

$$[X_\ell] = [\overline{X}] - [\mathcal{R}] - [\mathcal{S}_\ell] + [\mathcal{R} \cap \mathcal{S}_\ell] = \mathbb{L}^3 - \mathbb{L}.$$

In the tangent nonosculating case, the two lines coincide in the unique fibre over the tangency point p_0 and are distinct elsewhere. To make the resulting degeneration explicit, use the tangent normal form

$$L = aX + bY, \quad Q = cX^2 + dXY + eY^2, \quad \ell(LQ) = ad + bc.$$

Then $\mathcal{R} \cap \mathcal{S}_\ell$ is cut out by

$$ad + bc = 0, \quad a^2e - abd + b^2c = 0.$$

Set-theoretically it is the union of

$$\{a = c = 0\} \quad \text{and} \quad \{ad + bc = 0, ae - 2bd = 0\}.$$

The first component is the vertical $\mathbb{P}^1$ over $p_0 = [0 : 1]$; the second is the closure of the residual section. They meet only at $([a : b], [c : d : e]) = ([0 : 1], [0 : 0 : 1])$. Hence

$$[\mathcal{R} \cap \mathcal{S}_\ell] = 2[\mathbb{P}^1] - [\text{pt}] = 1 + 2\mathbb{L},$$

and inclusion–exclusion gives $[X_\ell] = \mathbb{L}^3$.

The osculating case is computed explicitly in Proposition 7.1; there $X_\ell \simeq \mathbb{G}_m \times \mathbb{A}^2$, so its class is $(\mathbb{L} - 1)\mathbb{L}^2 = \mathbb{L}^3 - \mathbb{L}^2$. $\square$

Corollary 5.2. *In the transverse case, $X_\ell \not\simeq \mathbb{A}^3$.*

Proof. Apply the compactly supported Hodge–Deligne polynomial $E_c : K_0(\text{Var}_{\mathbb{C}}) \rightarrow \mathbb{Z}[u, v]$. One obtains

$$E_c(X_\ell; u, v) = (uv)^3 - uv, \quad E_c(\mathbb{A}^3; u, v) = (uv)^3.$$

$\square$

The motivic equality $[X_\ell] = \mathbb{L}^3$ in the tangent case does not by itself imply $X_\ell \simeq \mathbb{A}^3$. We now prove the isomorphism.

6. THE TANGENT NONOSCULATING SLICE

Choose an affine coordinate on $\mathbb{P}^1$ in which the tangent hyperplane is the plane of depressed cubics. For finite roots p, q, r , its equation is

$$p + q + r = 0.$$

Write $\text{Sym}^2(\mathbb{P}^1) \simeq \mathbb{P}^2$. For $p \in \mathbb{P}^1$, let

$$L_p = \{D : p \in D\}, \quad H_p = \{D : p + D \in \mathcal{H}_\ell\}.$$

For finite p , the two lines meet at

$$s(p) = \{p, -2p\}.$$

At $p = \infty$ they coincide. The map

$$s : \mathbb{P}^1 \longrightarrow \text{Sym}^2(\mathbb{P}^1), \quad p \longmapsto \{p, -2p\},$$

has image a nonsingular conic C_φ . Write $\varphi(p) = -2p$ on the affine chart.

Define

$$X^* = \{(p, \Lambda) : s(p) \in \Lambda, \Lambda \neq L_p, H_p\},$$

where Λ is a line in $\mathbb{P}^2$.

Proposition 6.1. *Projection from the moving point $s(p)$ defines a Zariski-locally trivial $\mathbb{A}^1$ -bundle*

$$X_\ell \longrightarrow X^*.$$

Proof. A point of X_ℓ is a pair (p, D) with $D \notin L_p \cup H_p$. Since $s(p) = L_p \cap H_p$, one has $D \neq s(p)$, so the line $\Lambda = \overline{s(p)D}$ is well-defined and avoids the two forbidden lines. Conversely, for fixed $(p, \Lambda) \in X^*$, the point D may be any point of $\Lambda \setminus \{s(p)\} \simeq \mathbb{A}^1$. Globally this is the universal $\mathbb{P}^1$ -bundle of points on Λ with the section $D = s(p)$ removed. $\square$

Every line through $s(p)$ meets the conic C_φ at a second point, counted with multiplicity. Write that point as $s(q)$.

Proposition 6.2. *The second-intersection map*

$$\rho : X^* \longrightarrow C_\varphi \setminus \{s(\infty)\} \simeq \mathbb{A}^1, \quad (p, \Lambda) \longmapsto s(q),$$

is a Zariski-locally trivial $\mathbb{A}^1$ -bundle.

Proof. The forbidden line H_p is the chord joining $s(p)$ to $s(\infty)$, so $\Lambda \neq H_p$ is exactly the condition $q \neq \infty$. For fixed $q \in \mathbb{A}^1$, the point p determines the secant-tangent line through $s(p)$ and $s(q)$. This line equals L_p for exactly one value, $p = \varphi(q) = -2q$. Hence

$$\rho^{-1}(s(q)) \simeq \mathbb{P}^1 \setminus \{\varphi(q)\} \simeq \mathbb{A}^1.$$

The secant construction extends regularly across $p = q$ by taking the tangent line to the conic. At $q = 0$, that tangent line is L_0 ; the exceptional value is still the single point $p = \varphi(0) = 0$. Thus no extra fibre degeneration occurs. Equivalently, X^* is the complement of the graph of φ in the trivial $\mathbb{P}^1$ -bundle $\mathbb{P}^1 \times \mathbb{A}^1$, and is therefore an $\mathbb{A}^1$ -bundle. $\square$

Lemma 6.3. *Every Zariski-locally trivial $\mathbb{A}^1$ -bundle over $\mathbb{A}^n$ is trivial.*

Proof. Its transition maps have the form $t \mapsto a_{ij}t + b_{ij}$. The a_{ij} define a line bundle, which is trivial because $\text{Pic}(\mathbb{A}^n) = 0$. After choosing a trivialisation, the b_{ij} define a torsor under $\mathcal{O}_{\mathbb{A}^n}$, which is trivial because $H^1(\mathbb{A}^n, \mathcal{O}) = 0$. $\square$

Theorem 6.4. *If $\mathcal{H}_\ell$ is tangent but nonosculating to Γ , then $X_\ell \simeq \mathbb{A}^3$.*

Proof. By Proposition 6.2 and Lemma 6.3, one has $X^* \simeq \mathbb{A}^2$. Applying Proposition 6.1 and Lemma 6.3 then gives $X_\ell \simeq \mathbb{A}^3$. $\square$

Remark 6.5 (The hidden vector bundle). The moving point $s(p)$ is a conic in $\mathbb{P}(V_2)$. Its tautological line subbundle in $V_2 \otimes \mathcal{O}_{\mathbb{P}^1}$ is $\mathcal{O}(-2)$, and projection from $s(p)$ is governed by

$$0 \longrightarrow \mathcal{O}(-2) \longrightarrow \mathcal{O}^{\oplus 3} \longrightarrow \mathcal{O}(1)^{\oplus 2} \longrightarrow 0.$$

Projectivising the quotient gives $\mathbb{P}(\mathcal{O}(1)^{\oplus 2}) \simeq \mathbb{P}^1 \times \mathbb{P}^1$. Speyer wrote down the explicit maps in this sequence in the original public discussion [5].

7. THE OSCULATING SLICE

Proposition 7.1. *If $\mathcal{H}_\ell$ is osculating to Γ , then*

$$X_\ell \simeq \mathbb{G}_m \times \mathbb{A}^2.$$

Proof. Choose coordinates

$$L = aX + bY, \quad Q = cX^2 + dXY + eY^2$$

so that $\ell(LQ) = ac$. The slice equations are

$$ac = 1, \quad a^2e - abd + cb^2 = 1.$$

Thus $c = a^{-1}$ and

$$e = a^{-2} + a^{-1}bd - a^{-3}b^2.$$

The variables $a \in \mathbb{C}^\times$ and $b, d \in \mathbb{C}$ are free, giving

$$\mathcal{O}(X_\ell) \simeq \mathbb{C}[a, a^{-1}, b, d].$$

Hence $X_\ell \simeq \mathbb{G}_m \times \mathbb{A}^2$. In particular it has nonconstant units and cannot be $\mathbb{A}^3$. $\square$

Combining [Corollary 5.2](#), [Theorem 6.4](#), and [Proposition 7.1](#) proves [Theorem 1.2](#).

Corollary 7.2 (Essential uniqueness). *All successful linear–quadratic slices belong to a single natural orbit. More precisely, if $X_\ell \simeq \mathbb{A}^3$ and $X_{\ell'} \simeq \mathbb{A}^3$, then a change of variables in W and a rescaling of the functional induce an isomorphism of the two multiplication diagrams*

$$\begin{array}{ccc} X_\ell & \longrightarrow & H_\ell \\ \downarrow & & \downarrow \\ X_{\ell'} & \longrightarrow & H_{\ell'}. \end{array}$$

Thus, within this factorisation family, the resulting morphism is unique up to the natural linear symmetries. After identifying each source with $\mathbb{A}^3$, the corresponding Keller maps differ by a polynomial change of source coordinates and a linear change of target coordinates.

Proof. A tangent nonosculating hyperplane cuts Γ in a divisor $2p + q$ with $p \neq q$. The group PGL_2 acts transitively on ordered pairs of distinct points, and rescaling ℓ does not change the slice up to the normalization described above. $\square$

8. THE KELLER MAP AND ITS GLOBAL FIBRES

For a tangent representative, choose coordinates so that the affine hyperplane is

$$H = \{C = UX^3 + X^2Y + VXY^2 + WY^3\} \simeq \mathbb{A}_{U,V,W}^3.$$

By [Proposition 3.2](#), the associated map $f : X \rightarrow H$ has as fibre the simple roots of C . The fibre, image and nonproperness descriptions below also appear in the separately posted contemporaneous manuscript [\[2\]](#); we include a self-contained derivation in the normalized-slice framework.

Proposition 8.1. *The map f is étale of generic degree three. Its fibres have sizes 3, 1, 0 according as the cubic has three simple roots, one simple root, or no simple root. Its image is the complement of the triple-root curve*

$$\boxed{3UV = 1, \quad 27U^2W = 1.}$$

Over the discriminant complement, f is finite étale of degree three, and its nonproperness set is exactly the discriminant hypersurface.

Proof. Only the description of the triple-root curve requires calculation. If

$$UX^3 + X^2Y + VXY^2 + WY^3 = U(X - rY)^3,$$

comparison of the X^2Y coefficient gives $r = -1/(3U)$. Comparing the remaining coefficients gives $3UV = 1$ and $27U^2W = 1$. The fibre and image claims follow from [Proposition 3.2](#).

For properness, let

$$I = \{(C, p) \in H \times \mathbb{P}^1 : p \text{ is a root of } C\}.$$

The projection $I \rightarrow H$ is proper. Every fibre is finite because the coefficient of X^2Y is fixed and nonzero, so properness and quasi-finiteness make $I \rightarrow H$ finite. The other projection $I \rightarrow \mathbb{P}^1$ is an $\mathbb{A}^2$ -bundle: for each p , the root condition $C(p) = 0$ is a nonzero affine-linear equation in (U, V, W) . Thus I is irreducible. The simple-root locus in I is precisely X , and its complement is the multiple-root incidence. The image of that boundary is $V(\Delta)$, so f is finite over $H \setminus V(\Delta)$.

Conversely, take $y \in V(\Delta)$ and choose a boundary point $z \in (I \setminus X)_y$ marking a multiple root. Since I is irreducible and X is dense, algebraic curve selection gives a morphism

$$\gamma : \text{Spec } R \longrightarrow I$$

from a discrete valuation ring whose generic point maps to X and whose closed point maps to z . One may obtain R by taking the normalization of an integral curve through z that is not contained in $I \setminus X$ and localizing at a point over z .

If the restriction $f : X \rightarrow H$ were proper over some Zariski neighbourhood of y , the valuative criterion would extend the induced generic-point map $\text{Spec Frac}(R) \rightarrow X$ to $\text{Spec } R \rightarrow X$. After composing with the open immersion $X \hookrightarrow I$, this extension and γ would be two extensions over H of the same generic-point map. They must agree because the finite morphism $I \rightarrow H$ is separated, but their closed points would lie respectively in X and at $z \notin X$, a contradiction. Thus no neighbourhood of y supports a proper restriction. Together with finiteness away from the discriminant, this proves that the nonproperness set is exactly $V(\Delta)$. $\square$

The discriminant is

$$\Delta(U, V, W) = V^2 - 4UV^3 - 4W - 27U^2W^2 + 18UVW. \quad (3)$$

The source has no critical points. At a general point of $V(\Delta)$, the two markings of the coalescing roots escape to the deleted resultant divisor, while the marking of the remaining simple root stays finite. Thus the map misses only a curve, although it fails to be proper along an entire hypersurface.

Since $X \simeq \mathbb{A}^3$ by [Theorem 6.4](#), choosing polynomial coordinates on X and linear coordinates on H produces a polynomial self-map of $\mathbb{A}^3$ with nonzero constant Jacobian and generic degree three. The explicit coordinates in [Appendix A](#) identify it with the announced map and exhibit a three-point collision. Exact rational verification of the map and collision is also available in the public NASQRET notebook [\[4\]](#).

9. CONSEQUENCES AND LIMITS OF THE MECHANISM

9.1. Why degree three is exceptional. For the linear-times-degree- $(d - 1)$ family, the degree difference is $d - 2$. The local determinant, torsor degree and class-group determinant are therefore all controlled by the same integer:

$$\det D\Phi_{1,d-1} = \pm(d - 2) \text{Res}^2, \quad \deg q = d - 2, \quad \text{Cl}(U) \simeq \mathbb{Z}/(d - 2)\mathbb{Z}$$

when the boundary is prime. The cubic case is the unique one for which this integer equals one. The compactly supported Euler-characteristic argument in [Corollary 2.7](#) rules out the normalized affine slice for every $d \geq 4$, including reducible boundary cases.

9.2. Adjacent degrees and a next test case. The degree-difference obstruction disappears whenever $|s - r| = 1$. The first subsequent adjacent-degree case is $(r, s) = (2, 3)$, where the normalized slice has dimension five and generic degree

$$\binom{5}{2} = 10.$$

The present results do not decide whether a hyperplane slice can make that space $\mathbb{A}^5$. They reduce the plausible search space sharply: nonadjacent degrees are excluded by the cyclic torsor, while adjacent degrees require finer invariants analogous to the transverse Hodge–Deligne obstruction used here.

9.3. A multifactor obstruction. Let $k \geq 3$, let all d_i be positive, put $Y = \prod_{i=1}^k \mathbb{P}(V_{d_i})$, and set $D = \sum_i d_i$. A direct multifactor boundary complement removes every pairwise resultant divisor $\mathcal{R}_{ij}$ together with the pullback

$$\mathcal{S}_\ell = \{\ell(A_1 \cdots A_k) = 0\}$$

of a nonzero hyperplane in $\mathbb{P}(V_D)$. Each $\mathcal{R}_{ij}$ is prime, and the pairwise divisors are distinct: its class has entries d_j, d_i in the i, j positions of $\text{Pic}(Y) \simeq \mathbb{Z}^k$ and zero elsewhere.

Moreover, no $\mathcal{R}_{ij}$ is a component of $\mathcal{S}_\ell$. Otherwise $\ell(A_1 \cdots A_k)$ would vanish whenever A_i and A_j share a linear factor. Such products include every form $L^2 P$ with $L \in V_1$ and $P \in V_{D-2}$: over $\mathbb{C}$, factor P and distribute its linear factors among the remaining prescribed degrees. These forms span V_D . Indeed, taking $L = X$ gives all degree- D monomials whose Y -exponent is at most $D - 2$, and taking $L = Y$ gives those whose Y -exponent is at least 2. Thus ℓ would vanish identically, a contradiction.

Consequently the boundary has at least $\binom{k}{2} + 1 > k$ distinct prime components, whereas $\text{Pic}(Y)$ has rank k . The divisor localisation sequence therefore has a nonzero kernel in its boundary class map, producing a nonconstant unit on the complement. Hence this untrimmed multifactor boundary complement cannot be affine space. This unit obstruction does not assert the existence of a simultaneous normalisation of all pairwise resultants.

10. PROVENANCE AND SCOPE

The explicit Keller map was announced by Alpöge, who credited Fable for the construction [1]. In Speyer’s public discussion, Will Sawin posted a ChatGPT-produced symmetric-product formulation prompted by Andy Jiang. Sawin later developed an iterated $\mathbb{A}^1$ -bundle explanation through further ChatGPT discussion, and Daniel Litt supplied a related bundle argument [5]. Tao’s account gives the displayed parametrisation, inverse and induced map [6]. Levinson’s account gives the projective conic construction used in Section 6 [3]. Speyer supplied the exact vector-bundle sequence in Remark 6.5; a commenter using the name Skooi recorded the $d - 2$ divisor-class obstruction and a marked-simple-root description; and Shubhodip Mondal noted the corresponding $\mathbb{Z}/|r - s|\mathbb{Z}$ pattern for arbitrary pairs of binary factors [5]. Tao’s comment thread contains the explicit question whether the double-root orbit is uniquely characterized by polynomial triviality [6]. The fibre, image and nonproperness description in Proposition 8.1 also appears in the contemporaneous ulam.ai manuscript [2]; the argument here is the self-contained derivation presented in this manuscript.

The candidate contributions claimed by this manuscript are the general degree-difference determinant and torsor package, the fibre-functional criterion of Lemma 4.1, the motivic exclusion of the transverse orbit, the unit-theoretic identification of the osculating orbit, the complete orbit-by-orbit equivalence in Theorem 1.2, and the Euler-characteristic obstruction

for the normalized higher-degree slices. This description is not a priority or novelty determination; no exhaustive literature search has been performed. All statements in the main text are over $\mathbb{C}$ unless explicitly qualified.

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The supplementary SymPy program performs the finite list of exact checks described in the appendix. It does not encode the general geometric proofs. A passing run is local executable evidence for the listed formulas, not independent reproduction, peer review or end-to-end formalisation.

APPENDIX A. EXPLICIT AFFINE COORDINATES

Take the tangent representative

$$X = \left\{ (a, b, c, d, e) \in \mathbb{A}^5 : a^2e - abd + cb^2 = 1, \quad ad + bc = 1 \right\}.$$

The following formulas define a polynomial isomorphism $\Psi : \mathbb{A}_{a,y,z}^3 \rightarrow X$:

$$\begin{aligned} b &= 1 + ay, \\ c &= 1 - \frac{3}{2}ay + a^2z, \\ d &= \frac{1}{2}y - az + \frac{3}{2}ay^2 - a^2yz, \\ e &= -2z + 4y^2 - 4ayz + 3ay^3 - 2a^2y^2z. \end{aligned}$$

A polynomial inverse is

$$\begin{aligned} y &= 2bd - ae, \\ z &= 2d^2 + ce + 6bd^2 + 3bce - \frac{9}{2}e. \end{aligned}$$

These are the coordinates derived in Tao's account [6]. The identities

$$a(2bd - ae) = b - 1, \quad a^2(2d^2 + ce) = 2 + c - 3bc,$$

combined with the two defining equations give a short proof that the maps are inverse without localizing at $a \neq 0$.

Multiplication gives the three nonconstant coefficients

$$G(a, y, z) = (ac, \quad ae + bd, \quad be).$$

Expanded, they are

$$\begin{aligned} G_1 &= a - \frac{3}{2}a^2y + a^3z, \\ G_2 &= \frac{1}{2}y - 3az + 6ay^2 - 6a^2yz + \frac{9}{2}a^2y^3 - 3a^3y^2z, \\ G_3 &= -2z + 4y^2 - 6ayz + 7ay^3 - 6a^2y^2z + 3a^2y^4 - 2a^3y^3z. \end{aligned}$$

By [Theorem 2.2](#), or direct differentiation,

$$\det DG = -1.$$

Let

$$A_0(x, y, z) = \left(x, y, -\frac{1}{2}z\right), \quad B_0(u, v, w) = (w, 2v, 2u).$$

Then $F = B_0 \circ G \circ A_0$ is

$$\begin{aligned} F_1 &= (1 + xy)^3z + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2z + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2y - x^3z. \end{aligned}$$

It satisfies

$$\det DF = -2$$

and

$$F(0, 0, -1/4) = F(1, -3/2, 13/2) = F(-1, 3/2, 13/2) = (-1/4, 0, 0).$$

The target cubic at this collision has roots $\infty, 1/2, -1/2$ in the normalization used in the main text, so the three source points are exactly the three choices of marked simple root.

If (P, Q, R) denote the displayed output coordinates of F , then

$$U = \frac{R}{2}, \quad V = \frac{Q}{2}, \quad W = P.$$

Consequently the nonproperness hypersurface in the original coordinates is $V(D)$, where

$$D(P, Q, R) = 16P - Q^2 - 18PQR + Q^3R + 27P^2R^2 = -4\Delta\left(\frac{R}{2}, \frac{Q}{2}, P\right),$$

and the omitted triple-root curve is

$$3QR = 4, \quad 27PR^2 = 4.$$

APPENDIX B. EXACT VERIFICATION LEDGER

The supplementary script accompanying this paper performs the following checks over $\mathbb{Q}$ with symbolic assertions:

- (i) the four parametrisation formulas satisfy both defining equations of X identically;
- (ii) the displayed inverse formulas recover y and z identically;
- (iii) the expanded components of G equal $(ac, ae + bd, be)$ and $\det DG = -1$;
- (iv) $B_0 \circ G \circ A_0$ equals the announced map coefficient by coefficient, $\det DF = -2$, and the three rational points have the stated common image;
- (v) the discriminant of $UX^3 + X^2Y + VXY^2 + WY^3$ is [\(3\)](#);
- (vi) the triple-root equations and the collision-cubic factorisation hold exactly;
- (vii) for every positive pair with $r + s \leq 4$, the full resultant polynomial and the complete Jacobian determinant satisfy [Theorem 2.2](#) symbolically; and
- (viii) for every $1 \leq r, s \leq 8$, the Jacobian determinant of $\Phi_{r,s}$ at the normalising base point equals $(-1)^{s(r+1)}(r-s)$, including the zero base-point cases $r = s$.

No floating-point arithmetic or random testing is used.

REFERENCES

- [1] L. Alpöge, *Hello there the Jacobian conjecture is false*, public announcement on X, 20 July 2026.
- [2] Anonymous, *A counterexample to the Jacobian conjecture*, manuscript dated 20 July 2026.
- [3] J. Levinson, *The Jacobian conjecture counterexample, via classical projective geometry*, 22 July 2026.
- [4] NASQRET, *Exact SymPy verification — A Three-Dimensional Keller Counterexample*, 2026.
- [5] D. Speyer and contributors, *The new counterexample to the Jacobian conjecture*, Secret Blogging Seminar, 20–24 July 2026.
- [6] T. Tao, *A digestion of the Jacobian conjecture counterexample*, 21 July 2026, including comment thread.